

DUONG Quentin

Accelerator Physicist (PhD)
&
Nuclear Engineer

PROFILE



+33 (0)7.60.84.11.91



q.duong13@gmail.com



linkedin.com/in/duong-q



duongque.github.io



French driving licence

TOOLS & CODE

PYTHON



C / C++



VBA



POWERSHELL



BASH



LABVIEW



COMSOL



FLUKA



LANGUAGES



FRENCH – Native language



ENGLISH – C1



MANDARIN – A2



SPANISH – A2

HOBBIES



CINEMA



COOKING



MARTIAL ARTS



PHYSICS



WORK EXPERIENCE

R&D Engineer – Automation & Data

Jan 2026 – Today

Freelance (Paris – FRANCE)

Design of an end-to-end pipeline for processing unstructured documents (PDF): assisted information extraction, automated structuring and integration into a traceable data system.

➤ Python · Data pipeline · LLM API · Document automation · Traceability

Accelerator Physics Researcher

Oct 2022 – Sep 2025

CERN (Genève – SUISSE)

3-year doctoral program on the measurement and characterisation of electron cloud in the LHC. Regular work in radioactive environments with precision measurement instruments. Daily monitoring of machine operation and participation in CLEAR operations – a 200 MeV electron linac at CERN used for FLASH-VHEE research.

➤ Precision calibration · Instrumentation in radioactive environments · Data acquisition & analysis (7000+ fills) · PyECLOUD · Multiphysics simulation COMSOL · Monte Carlo radiation transport code FLUKA

Particle Physics Researcher

Apr 2022 – Aug 2022

KEK (Tsukuba – JAPAN)

5-month internship on cosmic muon measurements in an underground cavern with the COMET experiment at J-PARC. Muons used as reference particles for dosimetric calibration.

➤ Gaseous detector · Dosimetric calibration · Long-duration measurements (C++) · Advanced statistical analysis



EDUCATION

PhD | Highest Honours

Oct 2022 – Dec 2025

Paris-Saclay University (Paris – FRANCE)

Physics of Particle Accelerators

Thesis: Electron cloud study with the CERN LHC Vacuum Pilot Sector

➤ Particle accelerator technology · Proton beam dynamics · Vacuum systems · Electron-cloud dynamics · Beam-matter interactions · PyECLOUD · Machine Learning · Monte Carlo radiation transport code FLUKA

Master Degree | High Honours

Jan 2022 – Mar 2022

JUAS (Archamps – FRANCE)

Physics & Technology of Particle Accelerators

➤ Accelerator design · Beam dynamics & instrumentation · Collective effects · Normal & superconducting magnets · Synchrotron radiation · Cryogenics · RF Engineering · Vacuum systems

Oct 2020 – Dec 2021

Grenoble Alpes University (Grenoble – FRANCE)

Nuclear Physics

➤ Advanced nuclear physics · Plasma physics · Nuclear fusion

Engineer Degree | Honours

Oct 2019 – Aug 2022

Grenoble INP – PHELMIA (Grenoble – FRANCE)

Nuclear Engineering & Physics

➤ Reactor physics · Nuclear physics · Beam-matter interactions & Dosimetry · Radiation protection · Neutron transport simulation (Fortran, MCNP) · Thermohydraulics · Fluid mechanics

Bachelor Degree | Honours

Oct 2019 – Aug 2020

Grenoble INP – PHELMIA (Grenoble – FRANCE)

Physics, Electronics & Telecommunications

➤ Electronics · Signal treatment · Automatic control & Microprocessors · Material science · Network protocols · Computer science · Physics · Mathematics

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TRAINING

Topical course on Medical Accelerators

Jun 2026
CAS (Riga – LATVIA)

11-day intensive course organised by the CERN Accelerator School (CAS) in collaboration with Riga Technical University, covering medical applications of particle accelerators

➤ Medical accelerators · Radiobiology & Oncology · Radioisotope production · Nuclear Medicine · Medical imaging (PET/SPECT) · FLASH-VHEE · Radiotherapy · Regulatory framework

Challenge-Based Innovation on Particle Accelerators for Healthcare

Jun 2025 – Jul 2025
iFast (Archamps – FRANCE)

10 intensive days on medical innovation through particle accelerators. Complete technical proposal in a multidisciplinary team: "Fibrosis Disruption: SMITE Protocol for FLASH RT-Activated Anti-Fibrotic nanoparticles".

➤ Heavy ion radiobiology · Medical Linacs & Dosimetry · Radionuclide production · Accelerator physics & technology · Science mediation



PAPER & TALK

In situ estimation of maximum Secondary Electron Yield in the LHC Vacuum Pilot Sector beam pipe via electron cloud measurements and numerical simulations

Q. DUONG, V. BAGLIN, G. SATTONNAY

Accepted (Editors' Suggestion) – Physical Review Accelerator and Beams

Electron cloud energy distribution measurement and model in the LHC with the Vacuum Pilot Sector

Q. DUONG, V. BAGLIN, G. SATTONNAY

Under Reviewing – Physical Review Accelerator and Beams

SEY estimation with VPS electron cloud measurements and PyECLOUD simulations

CERN – TE/VSC Seminar
Apr 2025 – Meyrin, SWITZERLAND

Some bulk Cu and aC coating preliminary results obtained with the Vacuum Pilot Sector during 2023 run

Joint Accelerator Performance Workshop
Dec 2023 – Montreux, SWITZERLAND